

Xiaoqi Wu

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Currently based in: London / Suzhou

Data Analyst | Data Scientist | Shanghai / Suzhou / Hangzhou

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Programming and Statistics, Bioinformatics, Genetics, Cell Biology and Genetics, Applied Molecular Biology

Internship Experience

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Engineer · Analytical Department Shanghai 2024.07 - 2024.09

- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis on client-supplied samples, ensuring accuracy and reliability under strict quality-control procedures
- **Process Execution:** Conducted stability testing of drug samples using HPLC/UV, systematically identifying potential impurities and providing traceable data to support analytical conclusions
- **Quality Control:** Systematically identified potential impurities through rigorous quality-control procedures, providing key data to support product quality decisions – reflecting an engineering mindset toward data accuracy and traceability

Research & Project Experience

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery London 2025.12 - 2026.08

Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, bipartite record-matching evaluation, pytest

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers (spanning trait types, reporting formats, and data sources) from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set; reduced the 91-column VecTraits schema to a 22-field extraction schema aligned with the MIREVTD reporting standard
- **Multimodal Extraction Pipeline:** Built a multimodal evidence pipeline combining PyMuPDF full-text extraction, Docling table-structure parsing, and 150 DPI figure/table rendering, fed into Gemma 4 31B IT (a 256K-context multimodal LLM) to extract provenance-linked trait records from PDF text, tables, and images via prompt engineering and JSON Schema-constrained output; iterated across 11 prompt versions, with the final version reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Results:** Designed a bipartite record-matching evaluation framework (row-set, conditional-field, and end-to-end field metrics) with systematic error analysis; evaluated the locked pipeline on the held-out test set, achieving row precision of 1.000, row recall of 0.360, and conditional macro field F1 of 0.729 – confirming a "high-precision, structure-dependent recall" profile and supporting the pipeline's use as a candidate-record generator within a human-in-the-loop curation workflow

Modelling the Impact of Climate Change on *Aedes aegypti* Population Dynamics in Florida London 2025.03 - 2025.06

R, deSolve, ggplot2, ODEs, statistical validation (RMSE, Pearson, Spearman)

- **Model Adaptation:** Adapted a published tropical ODE compartment model to a subtropical geography to simulate *Aedes aegypti* population dynamics in Collier County, Florida under temperature and rainfall effects
- **Validation:** Validated the model against 53 weeks of local mosquito-surveillance data (RMSE=1.65, Pearson $r=0.29$, Spearman $\rho=0.30$), optimizing time-lag ($\tau = -2$ weeks) and a rainfall scaling factor

to quantify model performance

- **Outcome:** Proposed model improvements (anthropogenic water sources, spatial heterogeneity, uncertainty quantification), demonstrating end-to-end modelling and data-validation ability

Ecological Modelling and Population Dynamics Simulation

London 2024.01 - 2024.02

R, ggplot2, Lotka-Volterra, logistic growth model

- **Modelling Suite:** Built four population-dynamics models – metapopulation, fisheries harvesting (MSY), resource competition, and a 3-species predator-prey system, each implemented with difference equations and stochastic extensions; all four scored 90+ (up to 94), demonstrating solid algorithm implementation and parameter-optimization skills

Skills & Other

- **Python & Tooling:** Python (primary language), Git, virtual environments, Claude Code / Cursor for AI-assisted development; multi-backend LLM integration, prompt engineering, JSON Schema validation, pytest testing, async subprocess orchestration with JSON-based config/state tracking
- **Web Development:** FastAPI (routers/services architecture), React, TypeScript, Vite, Tailwind CSS
- **Data Analysis & Modelling:** R (ggplot2, dplyr, caper, deSolve), statistical inference (GLM, PGLS, Wilcoxon signed-rank, t-tests, Pearson/Spearman correlation), ODE-based dynamic system modelling and parameter optimization
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, skiing